As of: March 21, 2025
Received: March 15, 2025
Status: [redacted]
Tracking No. m8a-3w5z-rxj8
Comments Due: March 15, 2025
Submission Type: Web

PUBLIC SUBMISSION

Docket: NSF_FRDOC_0001
Recently Posted NSF Rules and Notices.

Comment On: NSF_FRDOC_0001-3479
Request for Information: Development of an Artificial Intelligence Action Plan

Document: NSF_FRDOC_0001-DRAFT-1768
Comment on FR Doc # 2025-02305

Submitter Information

Name: Charles Steiner
Email:

[redacted]

General Comment

I'm an AI researcher, and I think there's not enough attention on the problem that we could build powerful, impactful AI, and then suffer grave consequences because we didn't properly understand how to build it correctly.

Getting AI to do what you want is hard, particularly when "what you want" is complicated and hard to specify. As AI does more complicated real-world tasks for us, getting AI to do what you want (this is sometimes called "alignment," though unfortunately that word is used multiple ways) becomes a more and more valuable technical capability, both economically and strategically. American companies are currently the world leaders at it, and I want to see that continue.

But there's more than just the human vs. human contest here. There's human vs. nature. If we were talking about bridges, it's like American companies can build bridges that stay up 25% of the time, while Chinese bridges stay up 15% of the time. We're winning the human vs. human contest, but it does us a fat lot of good if we can't actually build good bridges.

Building AI that's smart like humans are is going to be a huge deal - it's like meeting an alien species. Because it's such a big deal, we don't want to make it go badly by building AI that doesn't do what we want. (Not out of malice, but out of not knowing how to do better. Like trying to build a bridge but it falls down.)

What roles do I think the government can play in accelerating American capabilities along this axis? (Taking it as understood that I'm neglecting other important axes of AI development.)

- Messaging may be more important than money. If the US government clearly communicates that it takes accident risks from AI seriously, and that we have to develop the technologies to surmount this human vs. nature contest, but that if we do so we can reap great rewards, others will follow our lead.

- A lot of people are going to answer this request for information by asking the US government to spend a lot of money on compute infrastructure. That won't help here. The bottleneck on the technologies needed to get AI to do what we want is understanding, not computer chips.

To that end, one way to spend money on this problem would be grant programs specifically for research into how to get AI to do what we want. In addition to direct work on improving the alignment of modern AI, indirect work is also needed to develop datasets and standards to evaluate progress, to build theoretical models of AI motivations, and to build tools for improving the alignment of future AI.

- Another role might be incentivizing AI labs to make progress on this research. One method could be brokering agreements between major AI labs and the US government in which the labs agree to do some fraction of research into getting AI to do what we want. Another method might be competitions or prizes for certain research - trying to create a "race to the top" rather than a race to the bottom (phrase taken from "Anthropic's Responsible Scaling Policy", anthropic.com, 2023), although government research would be needed to develop the standards for such competitions or prizes in the first place.

I've included my answers to what I think are some sensible questions a reader of this comment might have in an attachment, because they ran over the character limit.

Attachments

csteiner_comment_QA

Q: Why is getting AI to do what you want hard?
A: Because the current standard approach - training AI on whether humans approve or disapprove of its output (as in e.g. "Deepseek-V3 Technical Report", DeepSeek-AI, 2024. "Training a Helpful and Harmless Assistant with Reinforcement Learning from Human Feedback", Bai et al., 2022) - has fundamental problems handling human error. Models trained in this way will exploit flaws in human judgment to get greater approval (in fact LLMs already do mild versions of this, as in "Towards Understanding Sycophancy in Language Models", Sharma et al., 2023). There are more sophisticated approaches (e.g. "RLAIF vs. RLHF: Scaling Reinforcement Learning from Human Feedback with AI Feedback", Lee et al., 2023. "Deliberative alignment: reasoning enables safer language models", Guan et al. on openai.com, 2024. "Improving mathematical reasoning with process supervision", Lightman et al. on openai.com, 2023), but they still sometimes incentivize bad behavior (e.g. "Detecting misbehavior in frontier reasoning models", Baker et al. on openai.com, 2025). Finding better training methods is an open problem.

Q: It's all well and good to say AI should do "what we want," but what who wants, specifically?
A: Current AI to which this question applies (primarily LLMs) are built to follow common sense first, instructions from their creator organization second, and instructions from a normal user third. If AI is sufficiently useful and we continue with that current model, there may be a power struggle among humans for who gets to give instructions to that AI. Even those in current positions of power are individually unlikely to win in such a struggle. It therefore may be of strategic, not just moral, importance to build future AI to have a strong internal sense of morality inherited from a large sample of humanity, which it priviliges above user instructions, and to put other roadblocks in the path of aligning an AI with one person in particular. This strategic argument doesn't go all the way towards saying that we will end up building AI for the good of all humankind. Nevertheless, I am an idealist and would prefer if we really did build AI for the good of all humankind. The influence of the creator organization should be confined to the abstract questions needed to say what they mean by "the good of all humankind."

Q: Why is getting AI to do what you want economically and strategically valuable?
A: In the short term, it means that AI chatbots, or AI code assistants, or AI drivers, or AI factory optimizers will do good jobs by human standards. If you compare an AI driver that drives nicely 100% of the time, and an AI driver that drives nicely 95% of the time and confuses or scares the passengers the other 5%, the second one is not 95% as the first - it's much less. Improved AI alignment technology is valuable in the short term because it unlocks human-AI interaction that requires high reliability.

Q: Are we actually on track to build AI that's smart like humans are?
A: Yes. Not necessarily with the current generation of AI, but within a couple of decades. The best time to prepare is now.

Q: How would things go badly if we build a smart AI using the current approach to get it to do what we want?

A: It would learn to exploit systematic human errors to get reward (this already happens in "Deep reinforcement learning from human preferences", Christiano et al., 2017). One of the key things it will do is manipulate humans, but it may also hack computers, establish failsafes that are hard for humans to shut down, etc. In the worst case, this eventually leads to what a survey of AI experts categorized as "extremely bad outcomes (e.g. human extinction)." ("Thousands of AI Authors on the Future of AI", Grace et al. 2024)

Q: Will another government or international corporation try to build smart AI even if it doesn't do what they want? Or, put another way, "Will trying to build smart AI properly make us lose a human vs. human contest?"
A: Possibly. It's likely against their self-interest for reasons outlined above, but there are three sorts of reasons it might be done anyway:

1. The consequences of building different sorts of AI are a technical question about which some AI experts are in disagreement ("Thousands of AI Authors on the Future of AI", Grace et al. 2024). By chance, an actor may listen to particularly-unconcerned experts, and honestly believe they are pursuing their direct self-interest. Confidence may attract more investment money, leading to a selection bias among corporate actors.
2. If a person or faction within a government or corporation can advance their interests by having it run an AI project, the government/corporation may appear to act "irrationally" from the outside. These interests need not be financial. E.g. if Chinese internal politics heavily link AI to national pride, political careers may ride on promoting the appearance of a strong AI program without due care for the consequences.
3. A bad action may seem like the best option in context. If e.g. China fears that AI doing what Americans want is a threat to Chinese sovereignty, they may tolerate a higher AI accident risk (note that this overlaps with factor 2 - the government is an internal faction within the Chinese people). Or uncontrolled AI may be treated like a WMD, with actors attempting to possess it for deterrence or bargaining power.

Q: Will improving technology to get AI to do what we want have downsides? What if corporations stop their own research because they can just use public research? What if knowledge diffusion, including via stealing of secrets, helps other parties catch up?
A: Yes, retaining some secrets seems necessary to maintain our global lead, and allowing corporations to retain some secrets seems necessary to incentivize private research. We should also be maintaining our global lead in other ways (e.g. export controls, helping labs invest in cybersecurity to prevent stealing of secrets), and incentivizing private research in other ways (e.g. making agreements that include public commitments).

Q: How is this related to policy recommendations often coded as "AI Safety", but focused on AI being misused by malicious humans?
A: Most of them are irrelevant to this topic. This is not to say they're necessarily wrong - if AI being used by malicious humans does turn out to be a major problem, then we're going to have to make some hard choices about e.g. open source AI. But the problem this comment focuses on is AI

failing to do what we want, without any humans needing to be malicious. In this view, open source AI is primarily just a mild benefit to researchers.